Module 15 -Html

1. HTML Basics

Question: 1

Define HTML. What is the purpose of HTML in web development?

Answer:

HTML (HyperText Markup Language) is a standard markup language used to create and structure web pages.

- **Purpose of HTML in Web Development:**
  - To define the structure of a web page
  - To display content such as text, images, videos, and links
  - To instruct the browser how to present content
  - To work with CSS (for styling) and JavaScript (for functionality) to build complete websites

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Question: 2

Explain the basic structure of an HTML document. Identify the mandatory tags and their purposes.

Answer:

- **The basic structure of an HTML document is as follows:**

```
<!DOCTYPE html>
<html>
<head>
   <title>My Page</title>
</head>
<body>
   <h1>Hello World</h1>
</body>
</html>
```

- **Mandatory Tags and Their Purposes:**
  1. **<!DOCTYPE html>**

  → Declares the document type and HTML version (HTML5)

  2. **<html>**

  → Root element that contains the entire HTML document

3. **<head>**

→ Contains metadata like title, links, and styles

4. **<title>**

→ Sets the title of the webpage shown in the browser tab

5. **<body>**

→ Contains all the visible content of the webpage

Question: 3

What is the difference between block-level elements and inline elements in HTML? Provide examples of each.

Answer:

- **Block-level Elements:**
  - Take up the full width available
  - Always start on a new line
  - **Examples:**
    - <div>
    - <p>
    - <h1> to <h6>
- **Inline Elements:**
  - Take up only as much width as necessary
  - Do not start on a new line
  - **Examples:**
    - <span>
    - <a>
    - <img>
- **Difference (in short):**
  - **Block elements** → start on a new line
  - **Inline elements** → stay on the same line

Question: 4

Discuss the role of semantic HTML. Why is it important for accessibility and SEO? Provide examples of semantic elements.

Answer:

Semantic HTML uses meaningful tags that clearly describe the purpose of the content.

- **Examples of Semantic Elements:**
    - <header>
    - <footer>
    - <article>
    - <section>
    - <nav>

Importance:

1. **Accessibility:**
    - Helps screen readers understand content structure
    - Improves experience for users with disabilities
2. **SEO (Search Engine Optimization):**
    - Helps search engines understand the content better
    - Improves website ranking
3. **Code Readability:**
    - Makes code easier to read and maintain for developers

2. HTML Forms

Question: 1

What are HTML forms used for? Describe the purpose of the input, textarea, select, and button elements.

Answer:

HTML forms are used to collect user input and send that data to a server for processing. Forms are commonly used for login pages, registration, search, feedback, and more.

- **Form Elements and Their Purpose:**
  - **<input>**

    Used to take user input in different formats such as text, password, email, number, etc.

    Example: text field, password field

  - **<textarea>**

    Used to collect multi-line input from users

    Example: comments, messages

  - **<select>**

    Used to create a dropdown list for users to choose one option from multiple choices

  - **<button>**

    Used to submit the form or perform actions like reset or custom functions

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Question: 2

Explain the difference between the GET and POST methods in form submission. When should each be used?

Answer:

- **GET Method:**
  - Data is sent through the URL (visible in the address bar)
  - Limited data size
  - Less secure (data can be seen in URL)
  - Can be bookmarked
- **POST Method:**
  - Data is sent in the request body (not visible in URL)
  - Can send large amounts of data
  - More secure than GET
  - Cannot be bookmarked
- **When to Use:**
  - Use GET for retrieving data (e.g., search queries)
  - Use POST for sending sensitive or large data (e.g., login, registration forms)

Question: 3

What is the purpose of the label element in a form, and how does it improve accessibility?

Answer:

The **<label>** element is used to define a label for an input field in a form.

- **Purpose:**
  - It describes what the input field is for (e.g., Name, Email)
  - It is linked to the input field using the for attribute
- **Accessibility Benefits:**
  - Helps screen readers understand the form fields clearly
  - Improves usability for visually impaired users
  - Clicking on the label automatically focuses/selects the related input field
- **Example:**

```
<label for="name">Name:</label>

<input type="text" id="name">
```

3. HTML Tables

Question: 1

Explain the structure of an HTML table and the purpose of each of the following elements: <table>, <tr>, <th>, <td>, and <thead>.

Answer:

An HTML table is used to display data in rows and columns.

- **Table Structure Elements:**
  - **<table>**

    → Defines the entire table
  - **<tr> (Table Row)**

    → Represents a row in the table
  - **<th> (Table Header)**

    → Defines a header cell (usually bold and centered)
  - **<td> (Table Data)**

    → Defines a normal data cell
  - **<thead>**

    → Groups the header content of the table

- **Example:**

```
<table border="1">
  <thead>
    <tr>
      <th>Name</th>
      <th>Age</th>
    </tr>
  </thead>
  <tr>
    <td>John</td>
    <td>25</td>
  </tr>
</table>
```

Question: 2

What is the difference between colspan and rowspan in tables? Provide examples.

Answer:

- **colspan**

  → Used to merge columns (horizontal merge)

- **rowspan**

  → Used to merge rows (vertical merge)

- **Example:**

```
<table border="1">
  <tr>
    <th colspan="2">Student Info</th>
  </tr>
  <tr>
    <td rowspan="2">John</td>
    <td>Math</td>
  </tr>
  <tr>
    <td>Science</td>
  </tr>
</table>
```

Question: 3

Why should tables be used sparingly for layout purposes? What is a better alternative?

Answer:

Tables should be used only for displaying data, not for designing layouts.

- **Reasons:**
  - Makes code complex and harder to maintain
  - Not responsive for different screen sizes (mobile, tablet)
  - Poor accessibility for screen readers
  - Slower page rendering

- **Better Alternative:**
  - Use CSS for layout design
  - Techniques like:
    - Flexbox
    - CSS Grid
- These methods are more flexible, responsive, and easier to manage.